

MEDJUDGE: Medical Multimodal Reward Modeling

Yunhong He* Kai Zhang¹ Jiarong Qian² Zhengqing Yuan³
Yanfang Ye³ Jing Huang² Lichao Sun^{1*†}

¹Lehigh University ²Rice University

³University of Pennsylvania ⁴University of Notre Dame

Abstract

As large multimodal models (LMMs) become increasingly integrated into clinical workflows, robust evaluation and alignment mechanisms are critical to ensure their reliability in decision support. Existing reward models (RMs) typically reduce preference supervision to scalar scores, limiting transparency and diagnostic insight. In this work, we introduce **MEDJUDGE**, a multimodal medical reward modeling method that supports *interpretable*, *evidence-grounded*, and *clinically-aligned* decision evaluation. Given a clinical query (text and image) and candidate responses, MEDJUDGE outputs both strict pairwise preferences and concise rationales, enabling transparent verification and inference-time reranking. To overcome the lack of high-quality supervision data, we construct a large-scale synthetic dataset with five complementary generation heuristics, and compare three training paradigms—scalar (BT), supervised (SFT), and reasoning-augmented SFT (SFT-R)—across 8 open-source models. Our comprehensive experiments show that while all methods improve over baseline, SFT-R provides comparable accuracy with richer explanations, and BT achieves the strongest overall performance. Beyond decision accuracy, we propose UMLS-based Concept Overlap (UCO) to evaluate explanation quality, measuring concept-level alignment with clinician expectations. Compared to BERTScore, UCO better reflects human preferences across multiple statistical tests. We further show that our BT-trained judges improve Best-of-4 answer selection by up to 2.75%, demonstrating practical utility in clinical inference-time decision reranking. Our results establish MEDJUDGE as an effective, interpretable, and clinically grounded framework for multimodal medical reward modeling and evaluation.

*Yunhong He is an independent student researcher working remotely with Lichao Sun.

†Lichao Sun is the corresponding author.

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CCS Concepts

• Applied computing → Health informatics.

1 Introduction

Foundation models, especially large language models (LLMs) and multimodal LLMs, are increasingly explored and piloted as scientific and clinical copilots, supporting tasks ranging from visual question answering (VQA) [8, 11, 14, 16] to clinical documentation and workflow assistance [21, 40–42]. As these systems continue to develop, progress is bottlenecked not only by model capacity alone, but also by reliably evaluating, selecting and aligning model behavior with domain standards. **Reward Models (RMs)** have therefore become core infrastructure: they provide scalable evaluation, generate training signals for preference optimization, and enable inference-time search (e.g., Best-of-N) over competing solutions [20, 38].

The conventional RM is a discriminative scalar reward model: given an input x and response y , it predicts a single score $r_\theta(x, y)$. Training typically uses preference pairs and a Bradley–Terry (pairwise logistic, referred to as *BT*) objective, encouraging the preferred response to receive a higher score than the non-preferred one [3, 4, 20]. In practice, this corresponds to assigning numerical scores to candidate solutions and ranking them accordingly. However, scalar scoring is a limited interface: it is difficult to audit, provides little diagnostic signal about why one candidate is preferred, and underutilizes the generative capabilities of LMs that enable chain-of-thought and test-time scaling [18, 34, 43]. This has motivated generative reward modeling (GRM), where the “score” is represented as next-token probabilities (e.g., $p_\theta(\text{“A is better”}|x, y_A, y_B)$) optionally preceded by a rationale, enabling explicit reasoning and scalable verification via sampling and aggregation [28]. In parallel, RMs increasingly move toward rubric-based evaluation with explanations, aiming to make judgements transparent and criterion-consistent rather than a single scalar score.

In medicine, reward modeling has just recently received sustained attention, largely because curating high-quality preference data is expensive and requires substantial clinical expertise. Existing work has therefore progressed primarily in text-centric settings, including (i) domain-specific reward or preference modeling for clinical QA and radiology report generation [7, 19, 39], and (ii) process- or step-wise verification that evaluates intermediate reasoning to mitigate outcome-only supervision [1, 23, 38]. Despite these advances, medical multimodal verification remains substantially underexplored and no study has provided a systematic comparison of effective

training recipes. This gap is consequential because medical decision making is inherently multimodal, relying on heterogeneous imaging evidence (e.g., radiographs and histopathology) integrated with text history and examination findings [45]. In this regime, subtle visual errors can be clinically decisive and alter management, yet they are easy for both generators and evaluators to overlook [6, 37]. Moreover, existing evaluation commonly emphasizes decision-level agreement, leaving the alignment of clinical reasoning content under-measured.

To bridge this gap, we introduce **MEDJUDGE**, a multimodal medical evaluation method designed for medical decision-quality assessment. As shown in Fig. 1, given a clinical query (text + image) and two candidate responses, MEDJUDGE produces partial or complete outputs, including (i) a strict pairwise preference, (ii) a concise rationale grounded in the provided evidence. This interface supports inference-time selection and error analysis, while making verification transparent and clinically interpretable; moreover, the same structured outputs can serve as supervision for downstream alignment when desired.

Crucially, we target human alignment beyond decision accuracy. In addition to agreement with clinician preferences, we evaluate whether the judge’s justifications match the clinically salient concepts experts expect. We operationalize this via **UMLS-based Concept Overlap (UCO)**, a deterministic concept-level metric that maps text to normalized biomedical concepts and measures overlap, capturing clinical semantic alignment beyond surface similarity. Empirically, MEDJUDGE training improves not only preference agreement but also concept coverage and alignment, indicating more clinically grounded explanations. Our main contributions are threefold:

- **Data:** To overcome the scarcity of multimodal medical reward modeling data, particularly those requiring comparative judgments and clinically grounded rationales, we construct a synthetic dataset with five heuristics (structured, semantic-similarity, model-generated, perturbation, and random pairing). As shown in Table 3, removing any heuristic results in a consistent drop across all metrics (e.g., *UCO* drops by up to 0.0504 without structured data), validating the importance of data diversity for SFT-R training.
- **Methods & Evidence:** We compare four reward modeling strategies (Scalar, SFT, SFT-R, and BT) across 10 open-source models in Table 1. SFT improves performance over the baseline across all metrics, and SFT-R achieves comparable scores while offering richer explanations. For instance, GLM-4.1V improves from 47.87% (baseline) to 93.74% (SFT-R) in accuracy, and *UCO* increases from 0.1121 to 0.4016. This shows that reasoning enables more interpretable judgment without sacrificing quantitative reliability.
- **Human Alignment:** We introduce *UCO* (UMLS Concept Overlap) as a domain-specific explanation alignment metric and compare it with BERTScore using four statistical metrics in Figure 4. *UCO* outperforms BERTScore in BB:BG ratio (70.4% vs. 66.2%), Cohen’s *d* (0.6551 vs. 0.5540), and paired *t*- and *p*-values, confirming stronger alignment with human preferences. In Best-of-4 evaluation (Table 2), our BT-trained

judges further improve answer selection over no-ranking baselines (e.g., +2.75% on MedXpertQA-MM).

2 Related Work

2.1 Reward Models

Reward models (RM) can be characterized along two practical dimensions: reward formulation and scoring scheme [44]. Formulation is either *numeric-only*, produce a numeric reward for a query–response pair (e.g., a scalar score used for ranking or reinforcement learning) [4, 20] or *generative*, which producing natural language feedback (optionally with a final decision token) from which rewards or preferences can be derived [10, 43]. Scoring schemes are typically *absolute* (evaluate one candidate independently) or *comparative* (directly compare two candidates to express relative preference) [10]. Benchmarks such as RewardBench and JudgeBench highlight that RM quality is a nontrivial modeling target and can degrade under subtle errors or distribution shift [13, 25]. Medical reward modeling is comparatively recent and strongly constrained by expert labeling cost. As a result, prior work has concentrated on text-centric medical settings, including preference-based alignment for medical consultation and QA [7, 38, 39], as well as task-specific evaluators for radiology report generation (e.g., producing error analysis plus a score) [19]. Our work bridges this gap by introducing a multimodal medical RM that robustly supports heterogeneous imaging modalities.

2.2 Generative Reward Models

Generative reward models (GRMs) make the verification interface more transparent by producing structured feedback—for instance, a short rationale followed by a discrete decision token or rubric-scored criteria [10, 36, 43]. For example, GenRM formalizes verification as next-token prediction, enabling reasoning traces and aggregation (e.g., voting) for stronger verification [43]. RPM further emphasize deliberate reasoning as part of reward computation, using additional inference-time compute to improve reliability [10]. In multimodal settings, generalist critics such as LLaVA-Critic train large multimodal evaluators to judge diverse vision-language outputs [36], and unified verifier efforts aim to make verification more robust and reusable across tasks [17]. Recent evidence also suggests that matching the *final decision* alone may be insufficient: aligning the *reasoning process* can better discriminate RM reliability among strong models [27]. These trends motivate medical verifiers that move beyond opaque scoring toward structured, evidence-grounded judgments.

2.3 Inference-Time Scaling

RMs enable inference-time scaling by sampling multiple candidate solutions and selecting the best according to a verifier (Best-of-N), or by aggregating RM signals across samples [18, 35, 43]. Self-consistency is a representative decoding-time scaling approach that samples multiple reasoning paths and marginalizes to a more stable final answer [32]. Recent reasoning-oriented model families explicitly allocate more

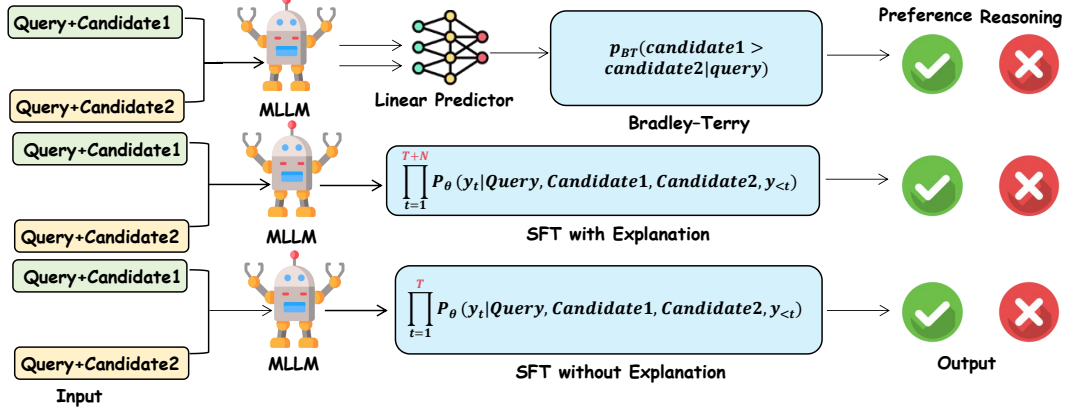


Figure 1: Comparison of three preference modeling paradigms for pairwise judgment: (top) scalar reward modeling with a Bradley–Terry objective, (middle) generative reward modeling trained via supervised fine-tuning (SFT) with explicit explanations, and (bottom) SFT without explanations.

computation at inference time (e.g., longer deliberation or richer reasoning traces) [9, 12], further increasing the importance of reliable RMs for candidate selection.

3 Methodology

We propose an end-to-end pipeline for medical preference modeling that jointly covers data synthesis and training paradigms. The pipeline synthesizes large-scale pairwise medical judgments with optional explanations using GPT-4o, and leverages the resulting supervision to train scalar and generative reward models under a shared multimodal backbone, enabling controlled study of preference reasoning and robust medical evaluation at inference time.

3.1 Training Paradigms

We study three representative reward modeling strategies that share the same multimodal backbone but differ in output form and the availability of explicit reasoning.

Scalar Reward Model with Bradley–Terry Objective. The conventional reward model learns a scalar scoring function $r_\theta(q, y) \in \mathbb{R}$, where q is the clinical query and y is a candidate response. Given a preference pair (y_1, y_2) , the preference probability is modeled using the Bradley–Terry formulation:

$$p_\theta(y_1 \succ y_2 | q) = \sigma(r_\theta(q, y_1) - r_\theta(q, y_2)), \quad (1)$$

with $\sigma(\cdot)$ denoting the sigmoid function. The model is trained by minimizing the negative log-likelihood of observed preference labels. This approach produces a single scalar score per candidate and does not provide explicit reasoning.

Generative Reward Model with Explanation. Instead of predicting scalar rewards, this model formulates preference judgment as a conditional generation task. Given (q, y_1, y_2) , the model generates a preference decision z together with an explanation. The likelihood is defined as

$$p_\theta(z | q, y_1, y_2) = \prod_{t=1}^T p_\theta(z_t | q, y_1, y_2, z_{<t}), \quad (2)$$

where the output sequence includes both reasoning and the final decision. Training is performed via SFT using ground-truth preference labels and clinically oriented rationales, enabling explicit reasoning and improved interpretability.

Generative Reward Model without Explanation. To isolate the effect of reasoning supervision, we consider a generative variant that predicts only the final preference decision. The conditional likelihood is

$$p_\theta(z | q, y_1, y_2) = \prod_{t=1}^{T'} p_\theta(z_t | q, y_1, y_2, z_{<t}), \quad (3)$$

where the output sequence is restricted to the decision token. This model retains the generative formulation while removing explicit explanation signals, serving as a controlled baseline.

3.2 MEDJUDGE Bench

To construct pairwise data for medical judgment, we build upon the dataset introduced in MedThink [8], which augments multiple public medical question-answering benchmarks, including PathVQA [11], RAD [14], and SLAKE [16], with explicit reasoning annotations. Following prior prompt-based data synthesis practices, we design five distinct but complementary prompts, each corresponding to a different heuristic for generating candidate responses in an open-ended medical judgment setting. Using these prompts, we initially construct 69,485 pairwise instances, where each instance consists of a reference response and a synthetically generated candidate response. We then apply strict format-based filtering to remove invalid or non-conforming pairs, resulting in a final dataset of 68,382 valid pairwise comparisons. Figure 2 summarizes the composition and balance of the synthesized dataset across answer preference, data sources, filtering stages, and synthesis heuristics. Let $\mathcal{D} = \{(x_i, r_i)\}_{i=1}^N$ denote the dataset, where x_i is an input medical query and r_i is its reference response. For each reference response r_i , we construct a candidate response c_i using one of the following five heuristics, forming a pair (r_i, c_i) for preference-based medical judgment.

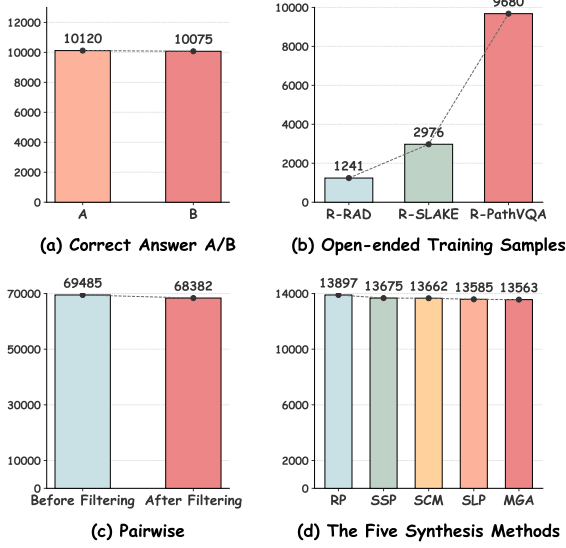


Figure 2: Distribution of the synthesized training data. (a) Distribution of correct answers between options A and B, showing a near-balanced preference split. (b) Number of open-ended training samples across different reward-oriented generation pipelines, including R-RAD, R-SLAKE, and R-PathVQA. (c) Statistics of pairwise preference samples before and after quality filtering, indicating minimal loss during data cleaning. (d) Sample counts generated by the five synthesis strategies (RP, SSP, SCM, SLP, MGA), demonstrating a well-balanced contribution from each method.

Random Pairing. Candidate responses are constructed by randomly sampling the response of a different input instance. For each reference response r_i , we uniformly sample an index $j \neq i$ and set $c_i = r_j$, $j \sim \text{Uniform}(\{1, \dots, N\} \setminus \{i\})$. This heuristic produces candidates that are clinically and contextually unrelated to the original query, introducing clear factual inconsistency and hallucination patterns.

Sentence-level Perturbation. Candidate responses are generated by applying sentence-level perturbations to the reference response. Given a reference response $r_i = [s_1, \dots, s_K]$ consisting of multiple sentences, we construct $c_i = \text{Perturb}(r_i)$, where $\text{Perturb}(\cdot)$ randomly removes, reorders, or partially modifies a subset of sentences.

Model-generated Alternatives. Candidate responses are also generated using a trained medical reasoning model conditioned on the same input. Specifically, $c_i \sim p_\theta(y \mid x_i)$, where p_θ denotes a pretrained or fine-tuned medical question-answering model, producing plausible but diverse alternatives.

Semantic Similarity Pairing. To construct challenging negative candidates, we pair reference responses with semantically similar responses from other instances. For each reference response r_i , we select $c_i = \arg \max_{r_j, j \neq i} \text{Sim}(r_i, r_j)$, where $\text{Sim}(\cdot, \cdot)$ denotes a sentence-level semantic similarity metric.

Structured Content Modification. Candidate responses are generated by selectively modifying medically relevant attributes in the reference response. Let $\mathcal{A}(r_i)$ denote a set of structured medical attributes extracted from r_i . We construct $c_i = \text{Modify}(r_i; \Delta\mathcal{A})$, where $\Delta\mathcal{A}$ represents controlled alterations to one or more attributes.

Data Split and Leakage Control. We randomly partition the filtered pairwise dataset into training and test sets with a ratio of 7:3. To prevent data leakage between the two splits, we perform exact-duplicate detection on the input queries and response pairs: any training instance that overlaps with the test set (i.e., shares an identical query or response pair) is removed from the training set. This ensures that the evaluation reflects the model’s generalization to unseen judgment instances rather than memorization of duplicated samples.

After constructing the pairwise response dataset, we further augment each pair (r_i, c_i) with a natural-language explanation that justifies the preference decision. Specifically, we query GPT-4o with a dedicated explanation prompt that presents the input x_i together with the two candidate responses and instructs the model to explain, in free-form text, why one response is preferable to the other from a medical perspective. Formally, for each pair (r_i, c_i) , we obtain $e_i = \text{Explain}(x_i, r_i, c_i)$, where e_i denotes the generated explanation. These explanations are not constrained to predefined error categories or scoring rules, and instead provide open-ended rationales that capture medical reasoning, factual consistency, and overall response quality. The generated explanations are used as auxiliary supervision signals during training, enabling MedJudge to learn both preference prediction and interpretable justification. As summarized in Figure 3, the explanation-augmented pairs are further organized into reasoning and no-reasoning training branches for different supervision objectives.

3.3 Prompt settings for synthetic data

In constructing the MedJudge training data, we adopt five complementary data synthesis strategies to generate pairwise comparison instances. Among them, four strategies are prompt-driven, where large language models are explicitly instructed to generate candidate responses according to predefined medical judgment heuristics (Random Pairing, Sentence-level Perturbation, Model-generated Alternatives, Semantic Similarity Pairing). These prompts are carefully designed to elicit systematic and interpretable variations in answer quality, enabling the judge model to learn fine-grained preference criteria.

4 Experiment

4.1 Implementation Details

We train MedJudge on a set of multimodal vision-language models with varying architectural designs and model scales, including Qwen3-VL [2] (4B and 8B variants), InternVL3.5-4B [30], GLM-4.1V-9B-Thinking [26], Skywork-VL-Reward-7B [31] and Unified Reward-2.0-qwen3vl-8b [33], which serve as representative baselines for medical judgment tasks. We further fine-tune MedJudge on two medical-domain models,

Table 1: Main evaluation results of multimodal reward modeling strategies across diverse model types. SFT denotes supervised fine-tuning, SFT-R denotes SFT with rationales, and BT denotes Bradley–Terry modeling.

Model	Setting	Accuracy(%)	Macro-F1(%)	AUC-ROC(%)	AP(%)	UCO(%)
General Model						
Qwen3-VL-4B-Thinking	Baseline	69.21	68.65	69.25	62.52	13.54
	SFT	90.54 (21.33↑)	91.29 (22.64↑)	91.41 (22.16↑)	87.55 (25.03↑)	-
	SFT-R	89.66 (20.45↑)	89.66 (21.01↑)	89.66 (20.41↑)	85.23 (22.71↑)	35.62 (22.08↑)
	BT	98.62 (29.41↑)	98.94 (30.29↑)	99.88 (30.63↑)	99.73 (37.21↑)	-
Qwen3-VL-8B-Thinking	Baseline	76.71	76.52	76.72	69.39	14.43
	SFT	91.34 (14.63↑)	90.26 (13.74↑)	90.01 (13.29↑)	90.01 (20.62↑)	-
	SFT-R	90.48 (13.77↑)	90.48 (13.96↑)	90.52 (13.80↑)	85.60 (16.21↑)	37.85 (23.42↑)
	BT	99.30 (22.59↑)	99.31 (22.79↑)	99.95 (23.23↑)	99.95 (30.56↑)	-
GLM-4.1V-9B-Thinking	Baseline	47.87	47.13	47.66	52.70	11.21
	SFT	96.67 (48.80↑)	95.14 (48.01↑)	95.80 (48.14↑)	92.99 (40.29↑)	-
	SFT-R	93.74 (45.87↑)	93.74 (46.61↑)	93.75 (46.09↑)	90.53 (37.83↑)	40.16 (28.95↑)
	BT	99.12 (51.25↑)	99.10 (51.97↑)	99.91 (52.25↑)	99.82 (47.12↑)	-
InternVL-3.5-4B	Baseline	60.55	57.10	60.58	55.99	12.98
	SFT	88.11 (27.56↑)	87.20 (30.10↑)	86.93 (26.35↑)	83.01 (27.02↑)	-
	SFT-R	86.54 (25.99↑)	86.42 (29.32↑)	86.41 (25.83↑)	83.63 (27.64↑)	32.11 (19.13↑)
	BT	98.00 (37.45↑)	98.13 (41.03↑)	99.00 (38.42↑)	99.10 (43.11↑)	-
Medical Model						
MedGemma-4B-it	Baseline	76.92	76.42	77.18	68.50	15.05
	SFT	92.81 (15.89↑)	92.61 (16.19↑)	90.99 (13.81↑)	87.13 (18.63↑)	-
	SFT-R	90.64 (13.72↑)	90.64 (14.22↑)	90.63 (13.45↑)	86.70 (18.20↑)	38.74 (23.69↑)
	BT	98.94 (22.02↑)	99.01 (22.59↑)	99.82 (22.64↑)	99.63 (31.13↑)	-
LLaVA-Med-1.5-7B	Baseline	47.61	45.22	46.71	49.27	11.56
	SFT	88.22 (40.61↑)	87.11 (41.89↑)	86.96 (40.25↑)	85.09 (35.82↑)	-
	SFT-R	81.22 (33.61↑)	80.09 (34.87↑)	81.53 (34.82↑)	80.41 (31.14↑)	29.41 (17.85↑)
	BT	90.24 (42.63↑)	89.73 (44.51↑)	92.02 (45.31↑)	90.13 (40.86↑)	-
Reward Model						
Skywork-VL-Reward-7B	Baseline	88.10	79.26	80.41	80.49	15.77
	SFT	99.11 (11.01↑)	99.09 (19.83↑)	99.89 (19.48↑)	99.91 (19.42↑)	-
	SFT-R	98.56 (10.46↑)	98.56 (19.30↑)	99.92 (19.51↑)	99.92 (19.43↑)	36.98 (21.21↑)
	BT	99.15 (11.05↑)	99.31 (20.05↑)	99.95 (19.54↑)	99.95 (19.46↑)	-
UnifiedReward-2.0-qwen3vl-8b	Baseline	53.74	49.11	50.20	51.00	11.13
	SFT	99.56 (45.82↑)	99.71 (50.60↑)	99.99 (49.79↑)	99.97 (48.97↑)	-
	SFT-R	99.14 (45.40↑)	99.21 (50.10↑)	99.89 (49.69↑)	99.05 (48.05↑)	44.42 (33.29↑)
	BT	99.30 (45.56↑)	99.31 (50.20↑)	99.95 (49.75↑)	99.95 (48.95↑)	-
Closed Source Model						
Gemini-2.5	<i>Zero-shot</i>	75.85	74.00	75.00	71.00	13.00
GPT-4o	<i>Zero-shot</i>	83.77	83.50	83.00	81.00	22.00

MedGemma-4B-IT [22] and LLaVA-Med-v1.5-Mistral-7B [15], to assess its effectiveness on domain-specialized backbones.

All models are fine-tuned using the LLaMA-Factory framework on 4× NVIDIA GH200 GPUs, each equipped with 96 GB of VRAM. Training is conducted with FlashAttention-2 [5] for efficient long-context attention computation, DeepSpeed ZeRO Stage 3 [29] for memory optimization, and the AdamW optimizer. The maximum input sequence length is set to 8192 tokens to support long-form clinical reasoning. The models are trained for 10 epochs in total, resulting in an overall training time of approximately 20 wall-clock hours.

4.2 Evaluation Metrics

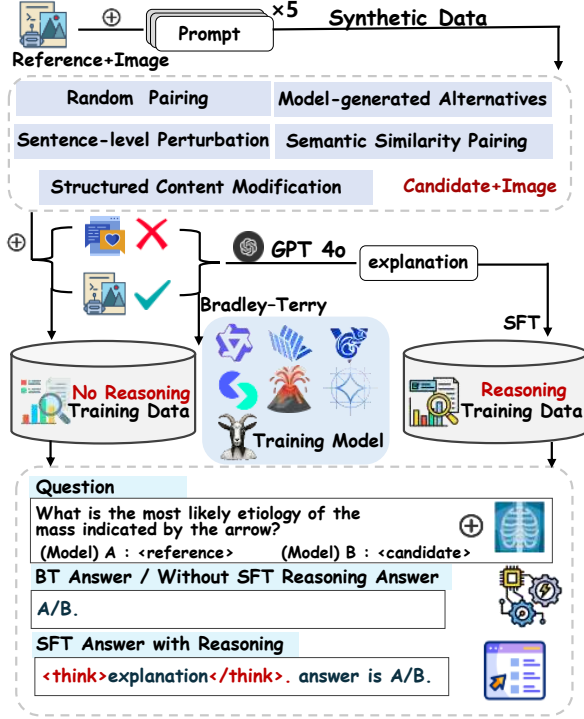
We evaluate MedJudge using multiple complementary metrics that capture both agreement with clinician preferences and the quality of ranking induced by the judge model, including Accuracy, Macro-F1, AUC-ROC, and Average Precision (AP). Detailed definitions and formulations of these standard metrics are provided in Appendix A.1.

$$\text{AUC} = \int_0^1 \text{TPR}(\text{FPR}) d\text{FPR}. \quad (4)$$

UMLS-based Concept Overlap (UCO) Metric. Our metric is inspired by recent LLM-based medical evaluation approaches such as GREEN, which assess factual correctness by prompting GPT-4 to identify and score clinically significant errors. While

Table 2: BoN (Best-of-4) accuracy across different judge models on three datasets. All values are in percentage (%) and measured with Qwen3-VL-4B as base model. Numbers in **red indicate improvements over the baseline (no ranking).**

Judge Model (BT Setting)	PathVQA (Close Ended)	VQA-RAD (Close Ended)	MedXpertQA-MM
Random candidate selection	61.91	68.47	20.85
GLM-4.1V-9B-Thinking	63.46 (+1.55)	69.49 (+1.02)	23.60 (+2.75)
Qwen3-VL-8B-Thinking	63.35 (+1.44)	69.44 (+0.97)	23.50 (+2.65)
Qwen3-VL-4B-Thinking	63.21 (+1.30)	69.30 (+0.83)	23.31 (+2.46)
InternVL-3.5-4B	62.84 (+0.93)	69.10 (+0.63)	22.85 (+2.00)
MedGemma-4B-it	62.95 (+1.04)	69.25 (+0.78)	23.10 (+2.25)
LLaVA-Med-1.5-7B	63.05 (+1.14)	69.20 (+0.73)	23.25 (+2.40)
Skywork-VL-Reward-7B	63.18 (+1.27)	69.33 (+0.86)	23.40 (+2.55)
UnifiedReward-2.0-qwen3vl-8b	63.28 (+1.37)	69.42 (+0.95)	23.50 (+2.65)

**Figure 3: Overview of the MedJudge data synthesis and training pipeline.**

effective, such methods rely on LLM-generated supervision and judgment, introducing a potential circularity where the same model family is involved in both generating evaluation signals and performing evaluation.

To address this limitation, we propose a concept-level evaluation metric that does not depend on generative LLM judgments. We normalize medical concepts by linking extracted entity mentions to UMLS CUIs.

For each mention e , the linker returns ranked candidate concepts with confidence scores, and we retain only the top

candidate if its score exceeds a threshold τ :

$$\text{link}(e) = \begin{cases} c_1, & s_1 \geq \tau, \\ \emptyset, & \text{otherwise.} \end{cases} \quad (5)$$

After removing duplicate CUIs, this yields two concept sets,

$$C^{gt} = \{c_1^{gt}, \dots, c_m^{gt}\}, \quad C^{pred} = \{c_1^{pred}, \dots, c_n^{pred}\}. \quad (6)$$

We define UCO as their Jaccard similarity:

$$\text{UCO} = \frac{|C^{gt} \cap C^{pred}|}{|C^{gt} \cup C^{pred}|}. \quad (7)$$

Implementation Details. We implement the biomedical NLP pipeline using scispaCy with `en_core_sci_sm` and its UMLS entity linker (`linker_name="umls"`), whose knowledge base is based on the **UMLS 2020AA** Metathesaurus. We disable abbreviation resolution, keep at most one concept per mention, set $\tau = 0.7$, and process documents in batches of 64. Before extraction, prediction and reference texts are lightly normalized by removing reasoning-trace markers such as `<think>` and collapsing redundant whitespace. The metric is deterministic given a fixed scispaCy model, UMLS release, and threshold, and does not use any generative model at evaluation time.

Handling Empty Concept Sets. If both C^{gt} and C^{pred} are empty, we define $\text{UCO} = 1$ by the convention $0/0 := 1$; if exactly one set is empty, $\text{UCO} = 0$. We also report concept-level precision, recall, and F_1 . For empty references, recall is defined as 1, while precision is 1 only when the prediction is also empty, and 0 otherwise.

4.3 Results

As shown in Table 1, every model improves after training—but the more informative story lies in *which* strategy wins on *which* axis. We read the table along three takeaways. First, the discriminative BT objective is the strongest ranker and classifier. Second, adding reasoning supervision (SFT-R) gives up a small amount of raw accuracy relative to BT in exchange for auditable rationales and markedly better biomedical grounding. Third, effective medical judgment does not require large-scale models: compact backbones, once given domain-specific

Table 3: Ablation study on synthetic data heuristics. Removing each data source individually from the SFT-R training stage leads to a performance drop, showing its contribution to final model capability.

Setting	Accuracy	Macro-F1	AUC-ROC	AP	UCO
Full (All)	93.74	93.74	0.9375	0.9053	0.4016
w/o Structured	91.25 (-2.49)	91.02 (-2.72)	0.9126 (-0.0249)	0.8710 (-0.0343)	0.3512 (-0.0504)
w/o Semantic Sim.	91.90 (-1.84)	91.35 (-2.39)	0.9158 (-0.0217)	0.8745 (-0.0308)	0.3631 (-0.0385)
w/o Model-gen.	92.25 (-1.49)	91.98 (-1.76)	0.9205 (-0.0170)	0.8851 (-0.0202)	0.3746 (-0.0270)
w/o Perturbation	92.95 (-0.79)	92.64 (-1.10)	0.9261 (-0.0114)	0.8923 (-0.0130)	0.3854 (-0.0162)
w/o Random Pairing	93.42 (-0.32)	93.25 (-0.49)	0.9306 (-0.0069)	0.8995 (-0.0058)	0.3925 (-0.0091)

training, become competitive judges that surpass much larger closed-source systems.

BT is the strongest ranker and classifier. On the metrics that directly measure how well a model separates preferred from dispreferred responses—accuracy, AUC-ROC, and AP—the Bradley–Terry objective is consistently the top performer. This is expected rather than incidental: BT optimizes a pairwise discriminative loss that matches the evaluation criterion directly, whereas the SFT-based variants learn judgment only implicitly through next-token prediction. The effect is large and uniform across backbones: GLM-4.1V-9B rises from 47.87% to 99.12% accuracy (Macro-F1 47.13% $\rightarrow$ 99.10%, over 50% absolute gain), and even the already reward-tuned UnifiedReward-2.0 climbs from 53.74% to 99.56%. Because BT exposes the model to explicit preference contrasts, it learns sharp decision boundaries, making it the method of choice whenever the downstream use is ranking or scoring.

Reasoning trades marginal accuracy for auditability and biomedical grounding. SFT-R trails BT by a small margin on accuracy and AUC, but it is the only variant that emits a natural-language justification for each decision. This makes its verdicts *auditable*—a property that matters in clinical settings, where a judgment without a rationale is difficult to trust or act on. More importantly, explanation supervision substantially improves UCO: GLM-4.1V increases from 0.1121 (baseline) to 0.4016, Qwen3-VL-8B from 0.1443 to 0.3785, and UnifiedReward reaches 0.4442. We attribute this to the supervision signal itself—requiring the model to articulate *why* one response is medically preferable forces it to ground decisions in biomedical concepts rather than surface cues, internalizing domain knowledge that a label-only objective leaves implicit. BT and SFT-R are therefore complementary rather than competing: BT for peak discriminative performance, SFT-R when transparency and biomedical consistency are required.

Effective medical judges do not require scale. Domain alignment matters more than parameter count. After BT training, MedGemma-4B-it reaches 98.94% accuracy and 0.9982 AUC-ROC, outperforming far larger closed-source generalists such as GPT-4o (83.77%) and Gemini-2.5 (75.85%); Qwen3-VL-4B similarly climbs from 69.21% to 98.62%. For specialized judgment, alignment to the target domain dominates raw scale—a practically important result, since compact judges are

cheaper to serve and easier to host on-premise where clinical data governance demands it. The same trend extends to models already optimized for reward modeling (Skywork-VL-Reward-7B, UnifiedReward-2.0), which improve further under our pipeline, indicating that the gains stem from our data and objectives rather than from any particular backbone.

4.4 Inference-Time Scaling

To assess the efficacy of inference-time re-ranking, we conduct a comprehensive comparison using eight different models as judge models, evaluating their ability to select the best answer among four candidates (**Best-of-4**) across three benchmarks: PathVQA (Close-ended Subset), VQA-RAD (Close-ended Subset), and MedXpertQA. As shown in Table 2, all judge models consistently improve performance over the baseline (i.e., random selection without re-ranking), with gains ranging from +0.63% to +2.75%. We observe that stronger judge models—such as **GLM-4.1V-9B** and **Qwen3-VL-8B-Thinking**—yield the most significant improvements. For instance, using GLM-4.1V-9B as the judge model leads to +1.55% and +1.02% improvements on PathVQA and VQA-RAD, respectively. On the more challenging MedXpertQA benchmark, the benefit is amplified to +2.75%, indicating that re-ranking is particularly helpful in low-resource or high-noise scenarios where model outputs vary more widely in quality. Lighter-weight judge models such as **InternVL-3.5-4B** and **MedGemma-4B-it** also deliver noticeable gains, though generally smaller than their larger counterparts. This suggests a trade-off between inference cost and re-ranking effectiveness, resource-constrained settings may still benefit from modest improvements with lighter judges.

Interestingly, reward-model-based judges like **Skywork-VL-Reward-7B** and **UnifiedReward-2.0** perform competitively, achieving improvements of +2.55% and +2.65% on MedXpertQA, respectively. These results demonstrate that preference-trained reward models, though optimized for reward prediction, can generalize well to selection scenarios during inference. Overall, these results highlight the generalizability and scalability of our inference-time ranking strategy. The magnitude of improvement correlates with the strength of the judge model, and the benefit is especially prominent in challenging settings like MedXpertQA. This supports our

Table 4: Effect of context length on SFT-R (Qwen3-VL-4B), showing how increasing the input context affects verification performance.

Context	Accuracy (%)	Macro-F1 (%)
8K	89.66	89.66
16K	91.43 (+1.77)	91.38 (+1.72)

design choice of incorporating a separate judge model to supervise inference-time selection.

4.5 Ablation Study

To train our judge model, we construct 68,382 preference-labeled pairs using five complementary heuristics: *Random Pairing*, *Sentence-level Perturbation*, *Model-generated Alternatives*, *Semantic Similarity Pairing*, and *Structured Content Modification*. These heuristics target different error types—ranging from factual inconsistency to subtle attribute-level mismatches.

Table 3 shows the performance of SFT-R models trained with one heuristic removed at a time. We find that **Structured Content Modification** is the most critical: its removal causes the largest drop across all metrics, including -2.49% accuracy and -0.0504 UCO. This highlights its unique role in teaching fine-grained clinical verification. Removing **Semantic Similarity Pairing** or **Model-generated Alternatives** also degrades performance notably, especially in AUC-ROC and AP, suggesting their value in modeling hard negatives and realistic distractors. In contrast, removing **Random Pairing** results in the smallest impact, indicating its main utility lies in anchoring obvious negative cases.

Because SFT-R emits long output sequences (rationale plus decision) and medical images expand into many visual tokens, the input budget can become a bottleneck. We compare an 8K and a 16K context window for SFT-R (Qwen3-VL-4B) in Table 4. Extending to 16K yields $+1.77\%$ accuracy and $+1.72\%$ Macro-F1: at 8K, visual and reasoning tokens compete for the limited budget and either fine-grained imaging detail or explanation quality is truncated, whereas 16K relaxes this constraint and lets the model encode both more completely.

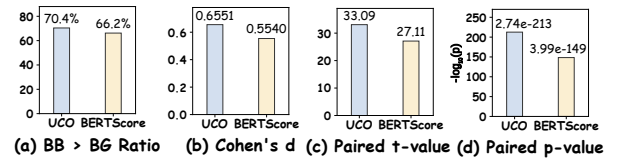
These findings validate the design of our synthetic supervision pipeline: both diverse error types and controlled perturbations are essential for training high-accuracy, well-calibrated judge models capable of response verification.

4.6 Consistency with Clinical Judgment

Our validation set is derived from a subset of MedXpertQA-MM [45], a multimodal dataset of clinical cases. We focus on the diagnosis subset. Each case contains patient history, symptoms, examination findings, at least one image, and a reference diagnostic label. Let q denote a medical question and $\{r_1, r_2, r_3\}$ denote model-generated reasoning answers for q . Based on physician annotations, two answers are considered to have comparable reasoning quality, while the remaining answer is judged inferior. For each comparison case, we compute

Table 5: Consistency between UCO and physician judgments on medical reasoning quality, evaluated on 4,143 physician-annotated comparison cases.

Metric	Value
UCO score (equivalent pairs, %)	16.02 ± 7.49
UCO score (pairs with inferior answer, %)	11.43 ± 6.48
Paired t -test (t)	33.09
Paired t -test (p)	2.74×10^{-213}
Wilcoxon test (p)	4.39×10^{-210}
Cohen’s d	0.66
UCO win rate (%)	70.40

**Figure 4: Comparative Statistical Analysis Between UCO and BERTScore.**

the UCO score between two physician-equivalent answers, denoted as $UCO(r_i, r_j)$, and between a physician-equivalent answer and an inferior answer, denoted as $UCO(r_i, r_k)$.

The evaluation is conducted on 4,143 physician-annotated comparison cases. If UCO is consistent with physician judgment, we expect $UCO(r_i, r_j) > UCO(r_i, r_k)$ to hold in the majority of cases. As shown in Table 5, the mean UCO score for physician-equivalent answer pairs is $16.02\% \pm 7.49\%$, while the score decreases to $11.43\% \pm 6.48\%$ when comparing a physician-equivalent answer with an inferior one.

The difference is statistically significant under a paired t -test with $t = 33.09$ and $p = 2.74 \times 10^{-213}$, as well as a one-sided Wilcoxon signed-rank test with $p = 4.39 \times 10^{-210}$, with an effect size of Cohen’s $d = 0.655$. Moreover, the inequality $UCO(r_i, r_j) > UCO(r_i, r_k)$ holds in **70.4%** of evaluated cases.

These results indicate that UCO is consistent with physician judgments in distinguishing answers of comparable medical reasoning quality from those judged to be inferior.

We further compare UCO with BERTScore under the same physician annotated comparison setting. Here, BB denotes cases where physicians judge the answers from model A and model B to be of comparable medical reasoning quality, while BG denotes cases where the answer from model A is judged to be clearly better than that from model C.

As shown in Figure 4, UCO exhibits a larger effect size (Cohen’s $d = 0.6551$ vs. 0.5540), a higher win rate (70.4% vs. 66.2%), and stronger paired t -test statistics ($t = 33.09$ vs. 27.11) compared to BERTScore under the same evaluation protocol. Both metrics achieve statistically significant results, while UCO shows clearer separation between BB and BG cases.

UCO deliberately measures concept *coverage*, not causal or logical validity; it complements rather than replaces full

Table 6: Reasoning-quality assessment of SFT-R (GLM-4.1V-9B) explanations on 100 randomly sampled test cases, rated against the reference explanation on a 3-point scale by an independent LLM evaluator.

Reasoning quality	Proportion
Correct	74%
Partially correct	19%
Incorrect	7%

Table 7: Disagreement analysis between BT and SFT-R (GLM-4.1V-9B) on the test set. On cases where the two paradigms disagree, BT is more often clinically correct, but the SFT-R rationale remains auditable.

Outcome on disagreement cases	Count (%)
Total disagreement cases	387
BT correct, SFT-R wrong	254 (65.6%)
SFT-R correct, BT wrong	98 (25.3%)
Both wrong	35 (9.0%)

Table 8: Out-of-domain evaluation on PathMMU (300 sampled instances, physician-validated labels), measured with GLM-4.1V-9B-Thinking.

Setting (GLM-4.1V-9B)	Accuracy (%)
Baseline	63.67
SFT-R	67.33 (+3.66)
BT	68.00 (+4.33)

reasoning evaluation. To probe whether high UCO co-occurs with correct reasoning, we use an independent LLM evaluator to rate the reasoning correctness of SFT-R (GLM-4.1V-9B) explanations on 100 randomly sampled test cases against the ground-truth reference, on a 3-point scale (Table 6). 74% of rationales are judged correct and 19% partially correct, indicating that the explanations are not merely naming the right concepts but largely produce logically valid reasoning. Automated rating is not equivalent to clinician review, but it provides a scalable, reproducible signal beyond UCO alone.

4.7 Out-of-Domain Generalization

To test whether the learned reward signal transfers beyond the synthetic construction pipeline, we evaluate on PathMMU [24], an external pathology benchmark with physician-validated ground-truth labels that is used in no stage of training. We randomly sample 300 instances and build pairwise preference pairs by pairing the correct answer with a randomly selected incorrect answer, then evaluate GLM-4.1V-9B-Thinking under three settings (Table 8). Both trained judges improve over the baseline (SFT-R +3.66%, BT +4.33%). The baseline accuracy

of 63.67% confirms that this is a genuinely harder, fully out-of-domain task than the in-domain evaluation in Table 1; the consistent gains indicate that the reward signal is not merely an artifact of the synthetic pipeline.

5 Limitations and Ethical Considerations

This work relies primarily on synthetically generated preference data, which may inherit biases or artifacts from the generating models despite the use of multiple complementary heuristics and ablation validation. Our evaluation focuses on multimodal medical QA and diagnostic reasoning, and does not cover the full complexity of real-world clinical workflows. The proposed UMLS-based Concept Overlap (UCO) metric depends on automatic entity extraction and linking, which may introduce noise for rare or ambiguous medical concepts. MedJudge is intended as a research tool for evaluation and alignment, not for autonomous clinical decision making. All experiments are conducted on publicly available or synthetic data, without the use of identifiable patient information.

While our out-of-domain (PathMMU) and physician-aligned (MedXpertQA-MM) evaluations provide evidence of transfer beyond synthetic supervision, they do not exhaust the complexity of real clinical workflows, uncertainty, conflicting image–text evidence, longitudinal context, and safety-critical edge cases. As the first systematic study of multimodal medical reward modeling, we view standardized-benchmark baselines as a necessary precursor to adversarial and deployment-grade evaluation, which we leave to future work.

6 Conclusion

We present MEDJUDGE, a unified and interpretable framework for multimodal medical reward modeling and evaluation. MEDJUDGE supports both scalar and generative preference modeling, produces structured justifications, and enables inference-time ranking for clinically grounded decision assessment. To address data scarcity, we synthesize a large-scale training dataset using five complementary heuristics, each of which contributes significantly to performance. Experiments across 10 open-source models show that all reward modeling strategies (SFT, SFT-R, and BT) yield substantial gains, with BT achieving the strongest quantitative performance and SFT-R providing richer explanations with higher biomedical concept alignment. We further introduce **UMLS-based Concept Overlap (UCO)**, a model-agnostic metric that better captures clinically meaningful explanation alignment than text-based metrics such as BERTScore. Under our supervised training, compact fine-tuned judges can exceed zero-shot proprietary reference systems. Overall, our results highlight the importance of transparent, explanation-aware reward modeling for reliable medical AI evaluation.

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A Appendix

A.1 Metrics

ACC. We evaluate MedJudge by measuring its agreement with clinician judgments. For each pairwise comparison, the judge model predicts which response is more clinically correct. A prediction is considered correct if it matches the aggregated clinician preference obtained via majority voting.

Formally, let y_i denote the clinician-preferred response for the i -th comparison and $\hat{y}_i$ be the model prediction. Accuracy is defined as

$$\text{ACC} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(\hat{y}_i = y_i). \quad (8)$$

Macro-F1. Macro-F1 evaluates how well the judge model identifies the preferred answer across different preference labels (e.g., A wins, B wins). The F1 score is first computed independently for each label based on the model’s predicted preference versus human annotations, and then averaged uniformly across all labels to avoid dominance by majority cases.

$$\text{F1}_c = \frac{2 \cdot \text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}, \quad (9)$$

$$\text{Macro-F1} = \frac{1}{C} \sum_{c=1}^C \text{F1}_c, \quad (10)$$

AUC-ROC. AUC-ROC measures our model’s ability to rank a preferred answer higher than a non-preferred one across all decision thresholds. Given a score difference $s = r(\text{cand}_1) - r(\text{cand}_2)$, AUC-ROC reflects the probability that the judge assigns a higher score to the human-preferred answer than to the rejected one. Since it is threshold-free, AUC-ROC captures the intrinsic discriminative power of the judge model rather than performance at a fixed cutoff, where the true positive rate and false positive rate are defined as

$$\text{TPR} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad \text{FPR} = \frac{\text{FP}}{\text{FP} + \text{TN}}. \quad (11)$$

AP (Average Precision, Judge Model). AP summarizes the Precision–Recall trade-off when ranking answer pairs by the judge model’s confidence (e.g., score gap or win probability). It emphasizes whether high-confidence judgments are correct, which is especially important when the judge is used at inference time to select the best answer from multiple candidates. AP can be interpreted as the average precision obtained as recall increases over ranked preference pairs:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad \text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \quad (12)$$

$$\text{AP} = \sum_n (R_n - R_{n-1}) P_n, \quad (13)$$

A.2 Qualitative Failure Modes

Figure 5 summarizes representative concerns raised from the perspective of external users when employing large language models as evaluators or judges in expert-oriented tasks. These

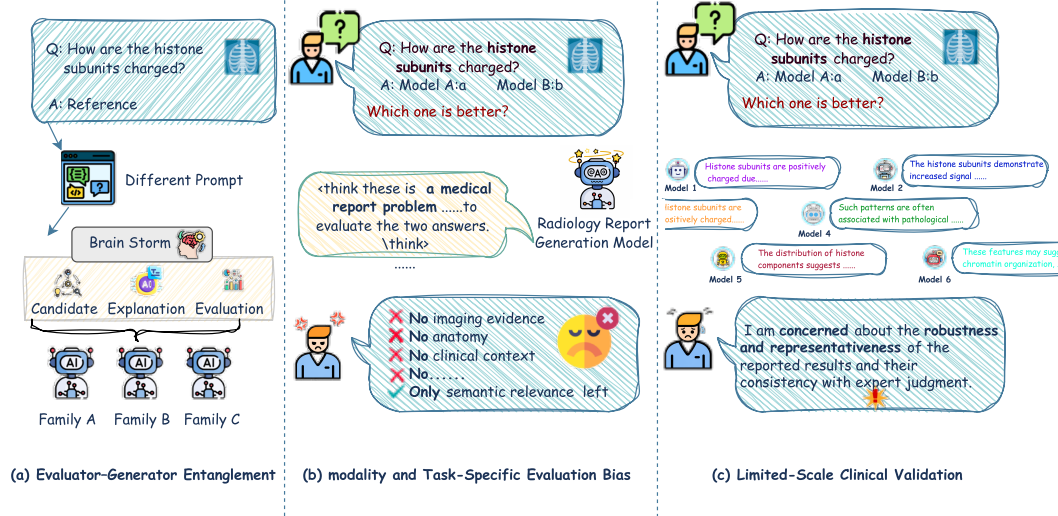


Figure 5: Challenges in evaluating medical reward models: (a) Prompt-induced entanglement, (b) modality and task-specific bias, (c) limited expert validation affecting robustness.



perspectives highlight practical limitations that may not be immediately apparent from aggregate benchmark scores alone.

First, users observe a potential **entanglement between evaluation and generation**. As illustrated in Figure 5(a), when multiple candidate explanations are produced under different prompts or reasoning styles, evaluation models may

implicitly favor certain linguistic patterns or explanation structures, rather than the underlying factual correctness. This raises concerns that evaluation outcomes may partially reflect prompt or generation artifacts instead of true task performance.

Second, users point out a **modality- and task-specific evaluation bias**. In expert domains such as radiology or clinical reporting (Figure 5(b)), evaluators are often required to judge answers without access to essential supporting evidence, including imaging data, anatomical localization, or clinical context. In such cases, the evaluation may collapse to surface-level semantic relevance, even though the task itself fundamentally depends on multimodal or domain-specific information.

Finally, users express concern regarding **limited-scale clinical validation and robustness**. As shown in Figure 5(c),

reported evaluation results may not fully reflect real-world expert judgment, especially when validation is conducted on a small number of cases or under controlled settings. This gap raises questions about consistency, representativeness, and the reliability of evaluation outcomes when deployed in practice.

Collectively, these user perspectives underscore the importance of cautious interpretation of automatic evaluation results and motivate the need for more robust, context-aware, and expert-aligned evaluation frameworks.